

codesign-mcdp: A Python Library for Monotone Co-Design Problems

Version 0.1.0 – Reference Manual

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Abstract

`codesign-mcdp` is a Python library for formulating and solving *Monotone Co-Design Problems* (MCDPs) in the framework of Censi (2015). A design problem is a relation between two posets, a functionality poset F and a resource poset R ; given a target functionality, the problem asks for the antichain of minimal resources needed to deliver it. Design problems compose under three operators (series, parallel, feedback), and the resulting class is closed under composition. The library implements the antichain calculus, six primitive design-problem types, the three composition operators, a Kleene fixed-point solver, two high-level builders (an MCDPL-style declarative builder and a modular `System` builder), and a collection of worked examples. This document is the reference manual for version 0.1.0.

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1 Introduction

A *co-design problem* is the joint design of components whose specifications mutually depend on each other. A canonical example: a battery must store enough energy to power an actuator, but the actuator must lift the battery itself, so the battery’s mass appears in the energy it has to supply. There is no way to fix one without fixing the other.

Censi (2015) showed that under modest monotonicity assumptions, such problems have a clean algebraic theory: the relations between resources and functionalities form a category closed

under series, parallel, and feedback composition, and solutions are computed by a Kleene fixed-point iteration in the lattice of antichains. The original publication included a software tool, MCDPL, distributed through `co-design.science`, which exposes the framework via a domain-specific language.

`codesign-mcdp` is a from-scratch Python implementation of the algorithmic core, designed around composable Python objects rather than a separate DSL. It targets users who want to formulate and solve MCDPs from inside their existing Python codebase without context switches to a DSL or external solver. The library is small (about 2000 lines of Python), has no required runtime dependencies beyond the standard library, and is licensed under MIT.

Scope of this document. This manual covers the public API and the mathematical structure it implements. Section 2 recalls the formal background. Sections 4 through 7 describe the data types, primitive DPs, composition operators, and the solver in detail. Sections 9 and 10 document the two high-level builders. Section 11 walks through every example shipped with the library. Section 12 collects modelling guidelines and common pitfalls. Section 13 discusses limitations and future directions.

Conventions. Code listings are typeset in monospaced font. Type signatures use Python’s standard notation ($f: F \rightarrow R$, for example). Mathematical quantities use $F, R, h, \top, \perp, \text{Min}$, etc.

2 Mathematical Background

This section recalls the formal definitions from Censi (2015) needed to specify the API. Proofs are omitted; the reader is referred to the paper for the full development.

2.1 Posets and Lattices

Definition 2.1 (Poset). A poset (*partially ordered set*) is a pair $(P, \leq)$ where $\leq$ is a reflexive, antisymmetric, transitive binary relation on the set P .

Definition 2.2 (Bottom and top). An element $\perp \in P$ is a bottom (*least element*) if $\perp \leq x$ for every $x \in P$. An element $\top \in P$ is a top (*greatest element*) if $x \leq \top$ for every $x \in P$. A poset has at most one bottom and at most one top.

Example 2.3 (Reals and Naturals). The non-negative reals augmented with infinity, $\overline{\mathbb{R}_{\geq 0}} = \mathbb{R}_{\geq 0} \cup \{+\infty\}$, ordered by the usual $\leq$, form a poset with $\perp = 0$ and $\top = +\infty$. Likewise $\overline{\mathbb{N}} = \mathbb{N} \cup \{+\infty\}$.

Example 2.4 (Product order). Given posets $(P_1, \leq_1), \dots, (P_n, \leq_n)$, the product $P_1 \times \dots \times P_n$ with the componentwise order

$$(x_1, \dots, x_n) \leq (y_1, \dots, y_n) \iff \forall i. x_i \leq_i y_i$$

is a poset. The bottom is $(\perp_1, \dots, \perp_n)$ and the top is $(\top_1, \dots, \top_n)$.

Definition 2.5 (Antichain). A subset $A \subseteq P$ is an antichain if no two distinct elements of A are comparable: $\forall x, y \in A, x \leq y \implies x = y$. We write $\mathbf{A}[P]$ for the set of all antichains of P .

Definition 2.6 (Min operator). For any subset $S \subseteq P$, the Min operator returns the set of minimal elements:

$$\text{Min}(S) = \{x \in S : \nexists y \in S. y < x\}.$$

$\text{Min}(S)$ is always an antichain. For a finite S , $\text{Min}(S)$ is the Pareto front of S .

2.2 Antichains as a Lattice

The set of antichains of a poset carries a natural order: $A \leq A'$ in $\mathbf{A}[P]$ iff every point in A' is dominated by some point in A . Intuitively, $A \leq A'$ means A is "lower" or "easier", having weaker Pareto-minimal demands.

Proposition 2.7. *($\mathbf{A}[P], \leq$) is a complete lattice when P is finite or chain-complete. Joins are computed by union followed by Min . The bottom of $\mathbf{A}[P]$ is $\{\perp_P\}$ (or $\emptyset$ if P has no bottom); the top is $\emptyset$.*

This lattice is the state space of the Kleene iteration (Section 6.3).

2.3 Design Problems

Definition 2.8 (Design Problem). *A design problem (DP) is a tuple $\langle F, R, h \rangle$ where F and R are posets and $h : F \rightarrow \mathbf{A}[R]$ is a function from functionality requests to antichains of resources. $h(f)$ is interpreted as the Pareto front of minimal resources required to deliver functionality f .*

Definition 2.9 (Monotone Design Problem). *A DP $\langle F, R, h \rangle$ is monotone (an MCDP) if h is monotone with respect to the antichain order: $f \leq f'$ in F implies $h(f) \leq h(f')$ in $\mathbf{A}[R]$.*

Intuitively, monotonicity says that demanding more functionality cannot reduce the minimal resources needed. This corresponds to the common-sense engineering principle that you don't get more for less.

2.4 Composition

The class of MCDPs is closed under three operators.

Definition 2.10 (Series). *Given DPs $d_1 = \langle F, M, h_1 \rangle$ and $d_2 = \langle M, R, h_2 \rangle$ whose interface poset M matches, their series composition is*

$$(d_1 \circ d_2)(f) = \text{Min} \left(\bigcup_{r_1 \in h_1(f)} h_2(r_1) \right).$$

Definition 2.11 (Parallel). *Given DPs $d_1 = \langle F_1, R_1, h_1 \rangle$ and $d_2 = \langle F_2, R_2, h_2 \rangle$, their parallel composition is $\langle F_1 \times F_2, R_1 \times R_2, h_1 \otimes h_2 \rangle$ where*

$$(h_1 \otimes h_2)(f_1, f_2) = \text{Min}(h_1(f_1) \times h_2(f_2)).$$

Definition 2.12 (Feedback). *Given a DP $d = \langle F \times X, R \times X, h \rangle$ with a common factor X , the feedback composition on X is the DP $d^{\dagger X} = \langle F, R, h^{\dagger X} \rangle$ where $h^{\dagger X}(f)$ is the antichain obtained by closing the recursion on X .*

The recursion is solved by Kleene iteration in $\mathbf{A}[R \times X]$, seeded at $\{\perp\}$ and ascending until a fixed point is reached.

2.5 The Kleene Fixed-Point Algorithm

Theorem 2.13 (Censi 2015, Prop. 4). *For an MCDP $d = \langle F \times X, R \times X, h \rangle$ closed on X , define the operator $\Phi_f : \mathbf{A}[R \times X] \rightarrow \mathbf{A}[R \times X]$ by*

$$\Phi_f(A) = \text{Min} \left(\bigcup_{r \in A} h(f, r_X) \cap \uparrow r \right),$$

where r_X is the X -component of r and $\uparrow r$ is the upward closure. Then Φ_f is monotone, and the Kleene ascent

$$A_0 = \{\perp\}, \quad A_{k+1} = \Phi_f(A_k)$$

converges to the least fixed point, which equals $h^{\dagger X}(f)$ projected to $\mathbf{A}[R]$.

In implementation, the iteration terminates when one of three conditions holds: $A_{k+1} = A_k$ (fixed point), $A_{k+1} = \emptyset$ (infeasibility), or every point in A_{k+1} has $r_X = \top_X$ (loop axis saturates).

3 Installation and Project Layout

`codesign-mcdp` requires Python 3.9 or newer. The package has no required runtime dependencies; `matplotlib` is needed only for example 05.

Installation. Clone the repository and install in editable mode:

```
1 git clone https://github.com/your-username/codesign-mcdp.git
2 cd codesign-mcdp
3 pip install -e .
```

Optional extras.

- `pip install -e ".[viz]"` adds `matplotlib` for the visualisation example.
- `pip install -e ".[nb]"` adds the Jupyter tooling needed to (re)generate the notebooks.

Repository layout.

```
codesign-mcdp/
  codesign/           the Python package (9 modules)
  examples/           eight runnable example scripts
  notebooks/          the same examples as executed .ipynb notebooks
  tests/              smoke tests
  docs/               this manual plus rendered trace images
  build_notebooks.py  script to regenerate the notebooks
  pyproject.toml      packaging metadata, PEP 621
  .github/workflows/  CI configuration
```

4 Core Data Types

This section documents the data types provided by `codesign.posets` and `codesign.antichains`.

4.1 Posets

The `Poset` abstract base class has the following interface:

```
1 class Poset(ABC):
2     name: str
3
4     def leq(self, x, y) → bool: ...      # is x ≤ y?
5     def eq(self, x, y) → bool: ...
6     def lt(self, x, y) → bool: ...
7     def bottom(self): ...                # least element, if any
8     def top(self): ...                   # greatest element, if any
9     def is_bottom(self, x) → bool: ...
10    def is_top(self, x) → bool: ...
```

```

11     def join(self, x, y): ...                                # least upper bound
12     def format(self, x) → str: ...

```

Four concrete poset classes are provided.

Reals(unit=""). The non-negative reals augmented with $+\infty$. Bottom is 0.0; top is `math.inf`. The optional `unit` string is used for pretty-printing only.

Naturals(unit=""). The non-negative integers augmented with $+\infty$. Bottom is 0; top is `math.inf`.

NamedProduct(components: dict[str, Poset]). The componentwise-ordered product of named factor posets. Elements are Python dicts mapping component names to values. The bottom is the dict of all bottoms; the top is the dict of all tops. **NamedProduct** can be nested freely: a component poset may itself be a **NamedProduct**.

Discrete(elements, leq=None). A finite poset with explicit element set and a caller-supplied `leq` predicate. Useful for catalog-like enumerations where the order is not the default discrete order.

4.2 Antichains

The **Antichain** class wraps a set of poset elements together with the poset they live in. The constructor normalises the input by removing dominated points and deduplicating.

```

1  class Antichain:
2      poset: Poset
3      points: tuple
4
5      @classmethod
6      def of_bottom(cls, poset) → "Antichain": ...
7      @classmethod
8      def empty(cls, poset) → "Antichain": ...
9      @classmethod
10     def singleton(cls, poset, point) → "Antichain": ...
11     @classmethod
12     def from_set(cls, poset, points) → "Antichain": ...
13
14     def union_min(self, other) → "Antichain": ...
15     def filter_above(self, floor) → "Antichain": ...
16     def leq(self, other) → bool: ...
17     def eq(self, other) → bool: ...
18     def has_any_top(self) → bool: ...
19     def is_empty(self) → bool: ...

```

The two key operations on antichains are `union_min`, which takes the union of two antichains and re-applies `Min`, and `filter_above`, which keeps only points dominating a given floor. Together they implement the body of the Kleene iteration in Theorem 2.13.

5 Primitive Design Problem Types

Six concrete subclasses of **DesignProblem** are provided, each appropriate for a different style of relation between functionality and resources.

5.1 AlgebraicDP

For relations where each resource is a closed-form monotone function of the functionality.

```
1 AlgebraicDP(  
2     F: NamedProduct,  
3     R: NamedProduct,  
4     equations: dict[str, Callable[[dict], float]],  
5     name: str = "algebraic",  
6 )
```

The `equations` dictionary maps each resource name to a callable taking a functionality dict and returning a scalar. `h(f)` returns a singleton antichain.

Example 5.1. *A battery sized by specific energy:*

```
1 battery = AlgebraicDP(  
2     F=NamedProduct({"capacity": Reals(unit="J")}),  
3     R=NamedProduct({"mass": Reals(unit="kg")}),  
4     equations={"mass": lambda f: f["capacity"] / 1.8e6},  
5 )
```

5.2 FunctionDP

For relations where the user supplies `h` directly as a function from functionality dicts to antichains. Use this when the relation is multi-valued (a true Pareto front) or has branching logic.

```
1 FunctionDP(  
2     F: NamedProduct,  
3     R: NamedProduct,  
4     h_fn: Callable[[dict], Antichain],  
5     name: str = "function",  
6 )
```

5.3 CatalogDP

For selecting from a finite catalog of implementations (motors, batteries, sensors).

```
1 CatalogDP(  
2     F: NamedProduct,  
3     R: NamedProduct,  
4     catalog: list[CatalogEntry],  
5     name: str = "catalog",  
6 )  
7  
8 CatalogEntry(  
9     name: str,  
10    provides: dict[str, Any],  
11    costs: dict[str, Any],  
12 )
```

For a request f , $h(f)$ is the antichain of `entry.costs` for every entry whose `provides` dominates f in F . Entries that are dominated by another feasible entry are pruned by `Min` in R .

5.4 ConstraintDP

A feasibility predicate plus a scalar cost, lifted to `Min`. Useful when the resource-functionality relation is most natural to express as *is this choice good enough, and how much does it cost?*.

```

1 ConstraintDP(
2     F: NamedProduct,
3     R: NamedProduct,
4     sampler: Callable[[dict], Iterable[dict]],
5     feasible: Callable[[dict, dict], bool],
6     cost: Callable[[dict, dict], dict],
7     name: str = "constraint",
8 )

```

The **sampler** yields candidate resource dicts for the given functionality; **feasible** filters them; **cost** maps the feasible ones into the resource poset, and the result is reduced by **Min**.

5.5 ODE_DP

Derives a monotone resource relation from a differential equation $\dot{x} = \text{rhs}(x, t, f)$. Two modes are supported.

Final value integrates the ODE from $t = 0$ to $t = t_{\text{end}}$ by explicit Euler and reports the extracted resources of the final state.

Steady state solves $\text{rhs}(x, \cdot, f) = 0$ for x by Newton iteration and reports the extracted resources of the steady-state value.

```

1 ODE_DP(
2     F: NamedProduct,
3     R: NamedProduct,
4     rhs: Callable,
5     extract: Callable[[state], dict],
6     mode: str,          # "final_value" or "steady_state"
7     t_end: float = 1.0,
8     n_steps: int = 100,
9     x0_fn: Callable[[dict], Any] = ...,
10    name: str = "ode",
11 )

```

5.6 UncertainDP

Brackets an unknown h with a known lower and upper bound. The optimistic mode (**lower**) yields a lower bound on the minimal resources; the pessimistic mode (**upper**) yields an upper bound. Section VII of Censi (2015) develops the theory.

```

1 UncertainDP(
2     F: NamedProduct,
3     R: NamedProduct,
4     lower: DesignProblem,
5     upper: DesignProblem,
6     mode: str = "upper",
7     name: str = "uncertain",
8 )

```

`with_mode("lower" | "upper")` returns the bracket selected for solving.

6 Composition Operators

6.1 Series


```

1 series(dp1: DesignProblem, dp2: DesignProblem) → Series
2 Series(dp1, dp2)

```

Requires $dp1.R == dp2.F$ structurally (same component names and posets). Implements the series definition: for each point of $h_1(f)$ run h_2 and take the union Min .

6.2 Parallel

```

1 par(dp1: DesignProblem, dp2: DesignProblem) → Parallel
2 Parallel(dp1, dp2)

```

Requires non-overlapping component names in F_1, F_2 and in R_1, R_2 . The output F and R are the concatenated `NamedProducts`. The antichain is the Cartesian product followed by Min .

6.3 Loop

```

1 loop(inner: DesignProblem, axis: str) → Loop
2 Loop(inner, axis="...")

```

Closes the feedback loop on a single named axis that must appear in both `inner.F` and `inner.R`. The resulting DP has $F = \text{inner.F}$ minus the axis and $R = \text{inner.R}$ minus the axis. Calling `h(f)` triggers the Kleene fixed-point iteration described in Theorem 2.13.

Remark 6.1 (Multiple feedback loops). *Loop closes one axis at a time. To close several loops, chain them: `loop(loop(inner, axis="x"), axis="y")`. The *System* builder (Section 10) closes all feedback loops simultaneously over a single bundled axis, which is usually more convenient.*

7 The Solver

7.1 Top-level entry point

```

1 solve(
2     dp: DesignProblem,
3     functionality: dict | None,
4     *,
5     max_iter: int = 200,
6     record_trace: bool = False,
7 ) → SolveResult

```

The result is a dataclass with the following fields:

```

1 @dataclass
2 class SolveResult:
3     antichain: Antichain      # the Pareto front (in dp.R)
4     iterations: int          # Kleene steps taken (0 if no loop)
5     converged: bool          # True if a fixed point was reached
6     feasible: bool           # True if antichain has finite, nonempty points
7     trace: list[Antichain]    # the sequence A_0, A_1, ... if recorded

```

7.2 The Kleene iteration

```

1 kleene_loop(
2     loop_dp: Loop,
3     f_outer: dict,
4     *,
5     max_iter: int = 200,

```

```

6     record_trace: bool = False,
7     trace_out: list | None = None,
8     info_out: dict | None = None,
9 ) → Antichain

```

The body of the iteration is direct from Theorem 2.13. The implementation adds:

- a *divergence cap* of 10^{30} that converts an iterate exceeding the cap to $\top$, so floating-point overflow becomes infeasibility rather than `inf/nan`;
- `try/except` around the inner h to catch `OverflowError`, `ValueError`, and `ZeroDivisionError`, again mapping to $\top$;
- three termination conditions: fixed point reached, antichain empty (infeasible), or every point's loop axis equal to $\top$ (provably infeasible).

7.3 Scalar minimisation over an antichain

```

1 minimize_cost(
2     result: SolveResult,
3     cost_fn: Callable[[dict], float],
4 ) → dict | None

```

Returns the single resource bundle in `result.antichain` that minimises `cost_fn`, or `None` if the result is infeasible. This is the scalarisation step that turns a Pareto front into a single engineering choice.

7.4 Observability: trace, verbose, callback, status

The solver can be made to report its progress in three independent ways, and its termination reason is exposed through a structured `status` field that is orthogonal to `feasible`.

The status field. `SolveResult.status` is one of three strings:

"converged" The iteration reached a fixed point (or provably hit $\top$) within `max_iter` steps. The answer in `result.antichain` is the algorithm's verdict; `feasible` tells you whether that verdict is finite (a real design) or $\top$ (no design exists).

"max_iter" The iteration was cut off at `max_iter`. The current antichain may or may not be near a fixed point. Usually a sign to increase `max_iter`, or to look for an unintentional non-monotonicity in the model.

"diverged" At least one numeric component crossed the divergence cap of 10^{30} before the iteration could settle, and the solver stopped. Distinguishes pure numerical blow-up from clean $\top$ -infeasibility.

The previous Boolean `converged` field is preserved as a backward-compatibility alias for `status == "converged"`.

Structured trace. Passing `trace=True` to `solve` populates `result.trace` with a list of `TraceEntry` dataclasses, one per Kleene step (plus the seed at iteration 0):

```

1 @dataclass
2 class TraceEntry:
3     iteration: int          # 0 = seed, then 1, 2, ...
4     antichain: Antichain    # snapshot at this step
5     n_points: int          # convenience: len(antichain)

```

```

6     delta:      float | None # max absolute change (numeric)
7                               # or 0/1 (discrete); None at iter 0
8     elapsed_ms: float        # wall time for this step alone

```

With `trace=False` (the default) the field is `None`, so the cost of unused tracing is zero.

Verbose printing. The integer `verbose` controls live output:

`verbose=0` (default) silent.

`verbose=1` one summary line at the end, e.g. `[solve] converged: 17 iters, |A|=1, total=2.0ms, feasible=True`.

`verbose=2` one line per iteration, showing the delta, the antichain size, and the step's wall time. Useful for watching oscillating fixed points.

Iteration callback. Passing `on_iteration=callable` registers a callback that receives each `TraceEntry` as soon as it is produced. Useful for live plots, custom logging, or interrupting an iteration that looks pathological:

```

1 def log_every_5(entry):
2     if entry.iteration % 5 == 0:
3         print(f"iter {entry.iteration}: delta={entry.delta:.3e}")
4
5 solve(dp, f, on_iteration=log_every_5)

```

8 Uncertainty

Two kinds of parameter uncertainty are supported, declared on `Module` instances and consumed by the same `solve` entry point. Both are independent of the model's structure: a module can have set-based uncertainty, stochastic uncertainty, both, or neither.

8.1 Declaration

A `Module` carries optional attributes:

```

1 class Battery(Module):
2     F = {"capacity": Reals(unit="J")}
3     R = {"mass":      Reals(unit="kg")}
4
5     def __init__(self, specific_energy=1.8e6, efficiency=0.85):
6         self.specific_energy = specific_energy
7         self.efficiency = efficiency
8         super().__init__()
9
10    def h(self, f):
11        return {"mass": f["capacity"]
12                / (self.specific_energy * self.efficiency)}
13
14    b = Battery()
15    b.uncertain_set = Box(...) # deterministic, set-based
16    b.uncertain_dist = Stochastic(...) # statistical

```

The uncertainty solver discovers every such module by walking the `_codesign_modules` attribute that `System.build` attaches to the returned DP. Before each solve, the module's named parameters are reassigned; afterwards, their nominal values are restored. The user's original `Battery()` instance is unchanged.

8.2 Set-based uncertainty (deterministic)

Box. Axis-aligned interval product. Each parameter is declared as (lo, hi) or (lo, hi, direction), where `direction` is one of the four tokens "more_is_better", "more_is_worse", "less_is_better", "less_is_worse". With all directions declared, the worst case is a single corner read off in constant time:

```
1 b.uncertain_set = Box(  
2     specific_energy=(1.6e6, 2.0e6, "more_is_better"),  
3     efficiency=(0.80, 0.90, "more_is_better"),  
4 )  
5 # worst case = (specific_energy=1.6e6, efficiency=0.80)
```

When some directions are undeclared, all 2^n endpoint combinations are evaluated and the worst is kept; cheap when n is modest.

Ellipsoid. An n -D ellipsoid $(p - c)^\top \Sigma^{-1} (p - c) \leq 1$ in parameter space. The covariance Σ encodes both scale and correlation between parameters, so an ellipsoid is a more honest model than a box when parameters are believed to vary together:

```
1 b.uncertain_set = Ellipsoid(  
2     center={"specific_energy": 1.8e6, "efficiency": 0.85},  
3     cov=[  
4         [1.0e10, -2.0e3],  
5         [-2.0e3, 2.5e-3],  
6     ],  
7     params=["specific_energy", "efficiency"],  
8     directions={  
9         "specific_energy": "more_is_better",  
10        "efficiency": "more_is_better",  
11    },  
12 )
```

With all directions declared, the worst case is the boundary point $c + Lu^*$, where L is the Cholesky factor of Σ and u^* is the unit vector in the direction of badness. Otherwise, the boundary is sampled (controlled by `boundary_samples`) and the worst sample is returned.

2D conveniences. `Disk(center, radius, ...)` and `Circle(center, radius, ...)` reduce to the isotropic ellipsoid $\Sigma = r^2 I$. For monotone systems with declared directions, the two are equivalent: the worst case lies on the boundary either way.

8.3 Stochastic uncertainty

Stochastic. A joint distribution built from named scipy-stats frozen marginals plus a copula:

```
1 from scipy import stats  
2  
3 b.uncertain_dist = Stochastic(  
4     marginals={  
5         "specific_energy": stats.uniform(loc=1.6e6, scale=0.4e6),  
6         "efficiency": stats.uniform(loc=0.80, scale=0.10),  
7     },  
8     copula=GaussianCopula(correlation=[[1.0, 0.4],  
9                                       [0.4, 1.0]]),  
10 )
```

Sampling is by the standard copula–marginal trick: the copula generates correlated uniforms in $[0, 1]^d$; the marginal ppfs map them to the named parameter axes.

Copulas. `Independence()` is the default (coordinatewise uniform). `GaussianCopula(correlation)` uses the Gaussian copula with the given correlation matrix; samples are drawn by Cholesky factorisation followed by Φ on each coordinate. Subclassing `Copula` requires only one method, `sample_uniform(n, d, rng)`, so other copulas (e.g. Clayton, Gumbel) drop in without further plumbing.

8.4 Querying

The same `solve` entry point accepts an `uncertainty` list and returns a richer object:

```

1 result = solve(
2     dp, f,
3     uncertainty=["worst_case", "mean", "p95", "cvar95", "samples"],
4     n_samples=1000,
5     rng_seed=42,
6 )
7
8 result.worst_case      # SolveResult at the worst point of the set
9 result.mean            # dict[r_port → mean across MC samples]
10 result.p95            # dict[r_port → 95th percentile]
11 result.cvar95         # dict[r_port → CVaR at 95% level]
12 result.samples        # list[Antichain], one per MC sample
13 result.feasibility_rate # fraction of feasible MC samples

```

The allowed labels are:

"worst_case" The deterministic worst over every module's `uncertain_set`. One canonical solve at the worst-case parameter point.

"mean", "p95", "cvar95" Marginal statistics over Monte Carlo samples, drawn from every module's `uncertain_dist`. The CVaR (conditional value at risk) at the 95% level is the mean of the worst 5% of samples.

"samples" The raw antichain per MC sample, for the user to aggregate however they like.

For a single solve call you typically observe the ordering

$$\text{nominal} < \text{mean} < \text{p95} < \text{CVaR95} < \text{worst_case},$$

which separates the four standard ways of reporting an answer "under uncertainty." Notebooks 11 and 12 visualise this for the example-7 drone.

9 The MCDP Builder

The `MCDP` class in `codesign.mcdpl` provides an MCDPL-style declarative syntax for a single self-contained design problem.

```

1 with MCDP("name") as m:
2     m.provides("functionality_name", unit="...") # outer F port
3     m.requires("resource_name", unit="...")     # outer R port
4
5     m.constraint("resource_name", lambda f: <expr>) # closed-form
6     # or
7     m.rule(lambda f: Antichain.from_set(...))      # multi-valued
8
9     m.loop_on("axis_name") # optional, may be repeated
10
11 dp = m.build()

```

The builder emits an `AlgebraicDP` or `FunctionDP` internally, then wraps it in `Loops` as requested by `loop_on`. Multiple `constraint` calls on the same target are joined (max). Mostly useful when the model has a small number of variables and reads naturally as a list of $\geq$ inequalities, mirroring the paper’s `mcdp { ... }` blocks.

10 The System Builder

For larger designs the `System` class in `codesign.system` provides modular composition with named subsystems and algebraic connection constraints. This is the recommended way to build non-trivial co-design models. Two equivalent surface syntaxes are available: the operator-overloaded form (recommended for new code) and the legacy lambda form (still supported).

10.1 Operator-overloaded syntax (recommended)

`provides`, `requires`, and `add` each return a port handle. Arithmetic operators on handles build expression trees lazily; the `>=` operator registers a constraint with the parent system.

```

1 from codesign import Module, Reals, System, solve
2
3 class Battery(Module):
4     F = {"capacity": Reals(unit="J")}
5     R = {"mass":      Reals(unit="kg")}
6     def h(self, f):
7         return {"mass": f["capacity"] / 1.8e6}
8
9 class Actuator(Module):
10    F = {"lift_force": Reals(unit="N")}
11    R = {"power":      Reals(unit="W")}
12    def h(self, f):
13        return {"power": 10.0 * f["lift_force"] ** 2}
14
15 sys = System("drone")
16 endurance      = sys.provides("endurance",    unit="s")
17 extra_payload  = sys.provides("extra_payload", unit="kg")
18 extra_power    = sys.provides("extra_power",   unit="W")
19 total_mass     = sys.requires("total_mass",    unit="kg")
20
21 battery = sys.add("battery", Battery())
22 actuator = sys.add("actuator", Actuator())
23
24 battery.capacity >= (actuator.power + extra_power) * endurance
25 actuator.lift_force >= 9.81 * (battery.mass + extra_payload)
26 total_mass >= battery.mass + extra_payload
27
28 drone = sys.build()

```

The four port kinds are:

- **Outer F** (from `provides`): can appear on the RHS of constraint expressions. Cannot be a constraint target.
- **Outer R** (from `requires`): can be a constraint target. Cannot appear in expression RHS.
- **Module F** (`handle.f_port`): can be a constraint target. Cannot appear in expression RHS (its value is determined by the constraints on it, not by the current iteration state).
- **Module R** (`handle.r_port`): can appear in expression RHS. Cannot be a constraint target (its value is determined by the module’s own `h`, not externally).

Misusing a port (e.g. trying to put a module F port on the RHS, or constraining an outer F) raises immediately with a message naming the port and explaining the rule. F and R port names within a single subsystem must be disjoint.

The supported operators are $+$, $-$, $*$, $/$, $**$, unary $-$, along with the helper functions `codesign.sqrt`, `exp`, and `log` for expressions that need transcendental terms. For demands that cannot be expressed algebraically, the legacy lambda form remains available and can be mixed freely with the operator form (see below).

10.2 Legacy lambda syntax

```

1 sys.constrain("battery.capacity",
2               lambda x: (x["actuator.power"] + x["extra_power"]) * x["endurance"])
3 sys.constrain("actuator.lift_force",
4               lambda x: 9.81 * (x["battery.mass"] + x["extra_payload"]))
5 sys.constrain("total_mass",
6               lambda x: x["battery.mass"] + x["extra_payload"])

```

The argument `x` is a dict carrying outer F values under their bare names (`x["endurance"]`) and subsystem R ports under their dotted names (`x["battery.mass"]`). Both syntaxes compile to the same internal representation; the operator form is purely sugar.

10.3 Class-based modules

The `Module` base class lets a design problem be defined declaratively:

```

1 class Battery(Module):
2     F = {"capacity": Reals(unit="J")}
3     R = {"mass":      Reals(unit="kg")}
4
5     def __init__(self, specific_energy=1.8e6):
6         self.specific_energy = specific_energy
7         super().__init__()
8
9     def h(self, f):
10        return {"mass": f["capacity"] / self.specific_energy}

```

A `Module` subclass declares its F and R ports as class-level dicts and overrides `h(self, f)`. The constructor wires everything into the underlying `DesignProblem` machinery. Parameterised modules are written by overriding `__init__` and calling `super().__init__()` after setting any instance attributes. `h` may return a dict (treated as a singleton antichain), a list of dicts (multi-valued antichain), or an `Antichain` directly.

10.4 Semantics

The `build` method emits a `Loop` whose axis bundles every subsystem's resource poset:

$$__modules__ = \prod_{m \in \text{modules}} R_m.$$

The inner DP performs one Kleene step:

1. Read the current bundle estimate to populate `ctx[m.r]` for every subsystem R port.
2. For each subsystem m , evaluate the constraint demands on its F ports to obtain f_m , then call `m.h(f_m)` to obtain an antichain.
3. Take the Cartesian product across subsystems; for each combination, evaluate the outer- R constraint demands to produce a candidate point.

4. Return the resulting antichain.

The outer Kleene iteration converges over all subsystem R ports simultaneously.

10.5 Validation

`build` performs the following checks:

- every subsystem F port has at least one `constrain` target;
- every outer R has at least one `constrain` target;
- every constraint target refers to a known module F port or a declared outer R ;
- module names do not clash with the reserved axis name `__modules__` and contain no `..`

If any check fails, `build` raises `ValueError` with a message identifying the missing or invalid item.

11 Worked Examples

The package ships twelve runnable examples under `examples/` and the same twelve as executed notebooks under `notebooks/`. Each is referenced below by its filename.

11.1 Example 1: the drone (Fig. 48 of the paper)

`01_drone.py`. The canonical MCDP from the paper. A battery powers an actuator that must lift the battery plus a payload, with extra fixed payload and avionics power. The model is a single `FunctionDP` closed by `Loop` on `battery_mass`. The example sweeps five mission profiles, three feasible and two infeasible. Iteration counts range from 8 to 41 depending on how close to the feasibility boundary the case is.

11.2 Example 2: integer optimisation (Sec. VI-D)

`02_integer_optimization.py`. The problem

$$x + y \geq \lceil \sqrt{x} \rceil + \lceil \sqrt{y} \rceil + c$$

over $\mathbb{N} \times \mathbb{N}$, parametrised by c . The inner DP enumerates all valid splits of the deficit, producing a multi-point antichain at each iteration. For $c = 4$ the iteration converges in six steps to $M(4) = \{(0, 7), (3, 6), (4, 4), (6, 3), (7, 0)\}$. The example notes that the paper’s claim $M(1) = \{(1, 0), (0, 1)\}$ contains a typo; the correct answer is $\{(0, 3), (3, 0)\}$, which is what the solver finds.

11.3 Example 3: AUV seabed surveying (Sec. VIII)

`03_auv_seabed.py`. An autonomous underwater vehicle sweeps an area A at velocity v with sensor radius r . Cyclic constraints couple time, energy, and cost. The example enumerates a small set of (v, r) tradeoffs per iteration, producing a three-point Pareto front per scenario.

11.4 Example 4: UncertainDP and ODE_DP

`04_uncertain_and_ode.py`. Two demos of the advanced primitives. First: a battery whose specific energy is known only to lie in $[1.6, 2.0]$ MJ/kg; the lower and upper brackets give optimistic and pessimistic masses. Second: a heater whose steady-state input power is recovered from Newton’s law of cooling via `ODE_DP`.

11.5 Example 5: visualising the Kleene ascent

`05_visualize_kleene.py`. Uses matplotlib to render the sequence $A_0, A_1, \dots$ for the Sec. VI-D problem, reproducing the structure of Fig. 36 in the paper. PNGs for $c \in \{1, 4, 8\}$ are committed under `docs/images/`.

11.6 Example 6: the drone in MCDPL syntax

`06_drone_mcdpl_syntax.py`. Rebuilds the drone with the **MCDP** builder. The model reads almost identically to the `mcdp { ... }` block in Fig. 48 of the paper. Output is identical to example 1.

11.7 Example 7: the drone, modular

`07_drone_modular.py`. Rebuilds the drone with the **System** builder. Battery and actuator are defined as **Module** subclasses with their own F and R , then wired together with three `>=` constraints in the operator-overloaded syntax. Fixed-point values match example 1 exactly; iteration counts are roughly $2\times$ because the bundle now has two coupled axes (battery mass, actuator power) instead of one.

11.8 Example 8: a modular vehicle with Pareto tradeoffs

`08_vehicle_modular.py`. Three independent subsystems: a **CatalogDP** motor (seven Pareto-incomparable entries), a linear chassis, and a battery. Five `>=` constraints close the loop on (motor, chassis, battery) resources. Two of the three mission profiles yield two-point system Pareto fronts (different motor choices trading total mass against total cost); the third is infeasible because no motor in the catalog can supply enough torque once the chassis grows to support the battery.

11.9 Example 9: a robotic arm with non-trivial topology

`09_robotic_arm.py`. Five subsystems (shoulder joint, elbow joint, sensor, controller, battery) wired with eight constraints. The connection graph is genuinely non-trivial: the controller's power demand depends on both the sensor rate and the joint command rate, the shoulder must support the elbow joint's mass at the end of its arm (introducing a mechanical cycle through `elbow.mass`), and the battery is sized by the integrated power of every electrical subsystem. The Kleene iteration resolves the resulting cycles in four steps for all three test mission profiles. This example is the cleanest demonstration of where the operator-overloaded syntax pays off: each constraint line states one physical relationship, and the cyclic structure emerges from the constraint graph rather than from explicit loop machinery.

11.10 Example 10: watching the solver work

`10_solver_trace.py`. The drone of example 7 instrumented to show every observability feature of `solve`: silent mode, the end-of-solve summary line (`verbose=1`), the per-iteration progress feed (`verbose=2`), the structured trace (`trace=True`), and a custom `on_iteration` callback. The same script runs the drone with three different `max_iter` caps and a deliberately under-resourced mission, demonstrating that `status` takes the three distinct values "converged", "max_iter", and "diverged" on these inputs. The notebook version (`10_solver_trace.ipynb`) adds a delta-vs-iteration plot showing the characteristic oscillating decay of the Kleene iteration to machine precision.

11.11 Example 11: set-based deterministic uncertainty

`11_uncertain_drone.py`. The drone's battery has two internal parameters (specific energy, efficiency) declared with "more is better" directions. First a `Box` with independent ranges, then a tilted `Ellipsoid` of the same widths. The worst-case mass under the box is +0.0158 kg above nominal; under the ellipsoid it is +0.0050 kg. The discrepancy is the cost of admitting the implausible corner where both parameters are simultaneously at their worst; the ellipsoid model rejects it explicitly.

11.12 Example 12: stochastic uncertainty with a Gaussian copula

`12_stochastic_drone.py`. The same drone with two uniform marginals tied by a Gaussian copula at correlation 0.4. A single `solve` call requests all five summaries: `worst_case`, `mean`, `p95`, `cvar95`, `samples`. Over a thousand Monte Carlo samples, the five values come out as nominal < mean < p95 < CVaR95 < worst_case, the canonical ordering. The notebook adds a matplotlib histogram of the MC distribution with vertical markers at each summary.

12 Modelling Guidelines and Pitfalls

A few patterns that come up repeatedly when authoring MCDPs.

Expose loop variables you want to inspect. When using the operator API directly, `Loop` projects its axis out of the outer R . To inspect the converged loop value, include it in the inner R under a *different name*: for example, both `battery_mass` as the loop axis and `report_mass` mirrored for outer- R visibility. The `System` builder handles this transparently.

Cap physical maxima. When a design variable has a physical ceiling ($v_{\max}$, $r_{\max}$), make h return a T-valued antichain whenever the iterate exceeds it. The Kleene ascent will then converge to infeasibility rather than oscillating or diverging.

Use `FunctionDP` or `CatalogDP` for breadth. `AlgebraicDP` always returns a singleton antichain. For a genuine Pareto front, use `FunctionDP` (enumerate the tradeoffs explicitly) or `CatalogDP` (provide multiple incomparable entries).

Scalarisation is downstream. The MCDP solver returns the Pareto front. Choosing which point to ship is a separate concern: apply `minimize_cost` with a scalar objective, or inspect the front and pick by hand.

Catalog Cartesian product can blow up. When several subsystems return multi-valued antichains, the inner h of the `System` takes their Cartesian product. For typical models (one or two catalog subsystems among many algebraic ones), this is not a bottleneck; but if many subsystems are catalog-driven, the inner antichain may grow large between `Min` prunings.

Numerical precision in feedback loops. `Antichain.eq` compares points exactly. For contractive floating-point feedback, the iteration may oscillate by an ulp near the fixed point and consume many iterations before converging. The divergence cap and `max_iter` guard against runaway, but choosing `max_iter` generously (200 to 500) is wise for nontrivial models.

13 Limitations and Future Work

Version 0.1.0 covers the algorithmic core: posets, antichains, six primitive DP types, three composition operators, the Kleene solver, and the two builders. The following are intentional omissions, all tractable extensions:

- **MCDPL surface syntax.** The paper’s `mcdp { ... }` text format is not parsed. The MCDP builder gives the same shape in Python.
- **Non-finitely-representable antichains.** The library works on finite, exactly-represented antichains. For relations whose Pareto front is continuous, the only support is the bracket pattern of `UncertainDP`. Approximation kernels (Sec. VII of Censi 2015) are not yet implemented.
- **Visualisation of the design graph.** Example 5 visualises the Kleene ascent over $\mathbb{N} \times \mathbb{N}$, but there is no general renderer for the operator tree or for the subsystem network of a `System`.
- **Symbolic/automatic differentiation of h .** All relations are evaluated numerically.
- **Distributed or parallel solving.** The Kleene iteration is sequential.

Contributions, particularly on these items, are welcome.

14 API Reference Summary

The public API exported by `import codesign` is summarised below.

Posets. `Reals`, `Naturals`, `NamedProduct`, `Discrete`, abstract base `Poset`.

Antichains. `Antichain`.

Design problems. `DesignProblem` (abstract base), `AlgebraicDP`, `FunctionDP`, `CatalogDP`, `CatalogEntry`, `ConstraintDP`, `ODE_DP`, `UncertainDP`.

Composition. `Series`, `Parallel`, `Loop` (classes); `series`, `par`, `loop` (function aliases).

Primitives. `adder`, `multiplier`, `scale`, `constant`, `identity`.

Solver. `solve`, `kleene_loop`, `minimize_cost`, `SolveResult`, `TraceEntry`.

Builders. `MCDP`, `System`.

Class-based modules. `Module` (declarative `DesignProblem` base class).

Constraint DSL. `Port`, `Expr`, `ModuleHandle` (returned by `System.add`), and the helper functions `sqrt`, `exp`, `log` for use inside constraint expressions.

Uncertainty. Set-based: `UncertaintySet` (abstract), `Box`, `Ellipsoid`, `Disk`, `Circle`. Stochastic: `Stochastic`, plus copulas `Independence` and `GaussianCopula`. Result: `UncertaintyResult`. Entry point: `solve_with_uncertainty` (the same machinery is also reachable through `solve(dp, f, uncertainty=[...])`).

Module-level. `__version__` (the string "0.1.0").

References

- [1] Andrea Censi. *A Mathematical Theory of Co-Design*. arXiv:1512.08055, 2015. <https://arxiv.org/abs/1512.08055>
- [2] B. A. Davey and H. A. Priestley. *Introduction to Lattices and Order*. Cambridge University Press, 2nd ed., 2002.
- [3] The MCDPL software distribution. <https://co-design.science/software/>